

# Data Modalities, Data Types, and the Data Explosion

CIS 635 — Knowledge Discovery & Data Mining

Course Reading Material

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*Unit: Foundations of Data — Modalities, Types, and Growth Trends*

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# 1 Learning Objectives

After completing this reading, you should be able to:

- Explain why data volume has grown exponentially in recent years, identify its major drivers, and explain why this growth motivated the emergence of data science as a distinct discipline.
- Distinguish structured data from semi-/unstructured data, give real-world examples of each, and explain why this grouping is a useful simplification rather than a rigid, universally agreed-upon taxonomy.
- Classify a given variable as numerical (discrete or continuous) or categorical (ordinal or nominal), and explain the practical consequences of misclassifying one as the other.
- Explain why raw modalities such as images, audio, and text must be encoded into numeric form before a data mining algorithm can process them.
- Compute the size of the numeric vector produced by encoding a binary, grayscale, or color image of a given resolution.
- Connect data modalities, data types, and data-volume growth to the broader KDD process introduced in the companion reading on knowledge discovery and data mining.

## 2 Introduction and Motivation: Why Data (and Data Science) Is Exploding

It is tempting to treat “data modalities” and “data types” as dry classification exercises — the kind of material that shows up on a quiz and then is never thought about again. They are not. Every decision a data mining system makes depends on first knowing *what kind of thing* a piece of data is and *how much* of it there is to work with. This reading builds that foundation, but it is worth starting with a question that is easy to take for granted: why does any of this matter *now*, more than it did twenty years ago? The answer is a simple, almost inconvenient fact: the amount of data in the world has grown so quickly, and so recently, that the tools and habits of mind built for smaller, tidier datasets no longer scale.

### 2.1 The Exponential Growth of Data

Figure 1 reproduces a widely cited estimate (Statista, 2024) [1] of the total volume of data created, captured, copied, and consumed worldwide each year, with actual figures through 2023 and forecasts through 2028.

Two features of this chart matter more than the exact numbers. First, the shape is not a straight line but a sharply upward curve: growth from 2010 to roughly 2018 is almost invisible on this scale, and the truly dramatic increase is concentrated in the last several years. Before 2010, the worldwide total was, in comparison to today’s figures, close enough to zero that it barely registers. Second, the units themselves are worth pausing on, because “zettabyte” is not a word most people use in daily life.

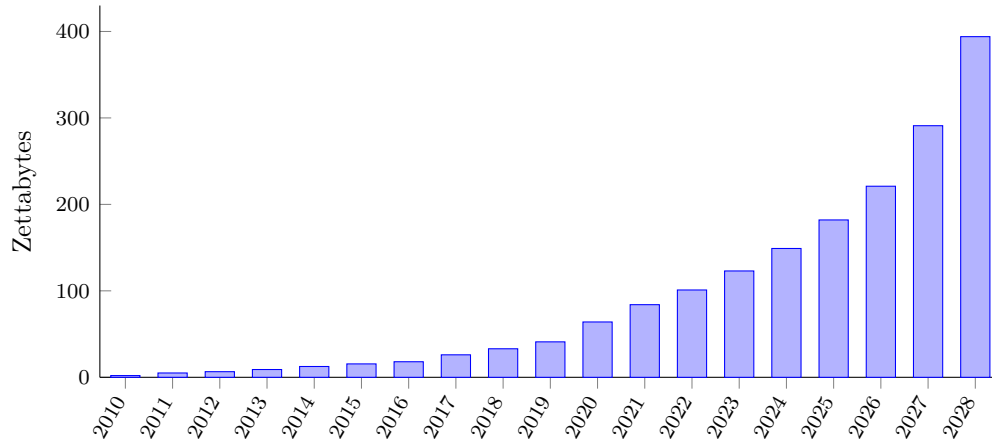


Figure 1: Volume of data created, captured, copied, and consumed worldwide, 2010–2023 (actual) with forecasts for 2024–2028, in zettabytes. Source: Statista (2024).

### Key Idea

Units of digital storage scale by factors of roughly 1,000: a **gigabyte (GB)** is about 1,000 megabytes; a **terabyte (TB)** is about 1,000 gigabytes; a **zettabyte (ZB)** is about 1 billion terabytes ( $10^{21}$  bytes). To make this concrete: a single terabyte is already enough to store roughly a quarter-million photos, and a large tech company’s *entire* global infrastructure might hold a few million terabytes. A zettabyte is a billion times that single-terabyte reference point — and the 2028 forecast in Figure 1 is 394 of them.

## 2.2 What Is Driving the Growth

This explosion did not happen because any single technology suddenly appeared; it happened because of a genuine shift in *how* data is created in the first place. For most of the twentieth century, everyday activities like taking a photograph or recording a voice memo produced **analog** artifacts — physical film, magnetic tape — that were not, by default, digital data at all. The shift to digital cameras, digital audio recorders, and eventually smartphones turned nearly every everyday activity into a source of digitally stored, transmittable, and analyzable data. This analog-to-digital transition, which felt gradual to the people living through it, is precisely what makes the last decade’s curve in Figure 1 look almost vertical: once an activity is digital by default, its data volume can grow as fast as the activity itself, with no physical medium acting as a natural bottleneck.

On top of that baseline shift, three more specific forces have accelerated growth further:

- **Social media.** Every post, comment, photo, video, like, and share is a data point, generated continuously by billions of users rather than periodically by a smaller number of institutions.
- **IoT (Internet of Things) sensors.** Thermostats, fitness trackers, industrial equipment, vehicles, and appliances now routinely report continuous streams of readings, a scale of automated data generation that has no analog-era precedent.
- **Generative AI and large language models.** The recent surge in tools such as ChatGPT is both a *consumer* of enormous training datasets and a *producer* of new synthetic data (generated text, images, and audio) at a pace that existing forecasts may still be underestimating.

### Tip

When you see a forecast like the 2024–2028 bars in Figure 1, treat it as a lower bound rather than a precise prediction. Forecasts built even a year or two before a major technological shift (such as the mainstream adoption of generative AI) tend to understate what actually happens, simply because the forecasters could not have modeled a driver that did not yet exist at consumer scale.

## 2.3 Why This Matters: The Rise of Data Science

Confronted with this volume and pace of data, organizations face a genuinely new problem: not a shortage of data, but an inability to make sense of the data they already have using manual review or traditional, hand-built reporting alone. This is precisely the gap that **data science** — and, within it, knowledge discovery and data mining, the subject of this course — has emerged to fill. It is worth remembering, though, that almost none of the underlying mathematics is new. The probability and statistics foundations this field depends on trace back centuries: Gerolamo Cardano’s early work on probability in games of chance, Jacob Bernoulli’s law of large numbers, Thomas Bayes’s framework for updating beliefs given evidence, Carl Friedrich Gauss’s contributions to probabilistic error analysis and the normal distribution, and Andrey Markov’s theory of sequential, state-dependent processes. What has changed is not the mathematics but the *volume and variety* of data these ideas are now applied to, and the computational power available to apply them at scale.

### Key Idea

Data science did not invent statistics; it *industrialized* it. The tools this course teaches — structured processes for cleaning, transforming, and mining data, discussed at length in the companion KDD-process reading — exist specifically because centuries-old statistical reasoning, applied by hand to small samples, cannot keep pace with data arriving at the scale shown in Figure 1.

### Practical Exercises

1. Using the unit conversions in this section, estimate roughly how many terabytes correspond to the 2028 forecast of 394 zettabytes. What does this number suggest about why individual organizations, rather than any single computer, must coordinate to store and process data at this scale?
2. Pick one of the three growth drivers (social media, IoT sensors, generative AI). Describe one concrete way it generates data that would not have existed, in any form, twenty-five years ago.
3. The reading argues that forecasts of future data volume are more likely to underestimate than overestimate. Propose one additional driver of data growth, not mentioned in this section, that a 2024 forecast might not have fully accounted for.
4. Explain, in your own words, why the analog-to-digital shift (film cameras to digital cameras, tape recorders to digital audio) is a necessary precondition for the kind of exponential growth shown in Figure 1, rather than merely a coincidental, unrelated trend.

## 3 The Landscape of Everyday Data: Data Modalities

Granting that enormous quantities of data now exist, the next natural question is: *what does all of it actually look like?* This section surveys the everyday data most people generate or encounter without necessarily thinking of it as “data” at all — a bank statement, a photograph, a search history, a doctor’s note — and introduces the vocabulary this course uses to group it.

### 3.1 Why a Clean Taxonomy Is Hard

A reasonable first instinct is to look for a small number of clearly separated categories that every piece of data falls into, with no overlap. In practice, this turns out to be surprisingly difficult. Data arrives in different **formats** (a spreadsheet row, a JPEG file, an audio waveform, a block of free text) and different **modalities** (numbers, images, sound, language), and a single real-world entity — a hospital patient, a bank customer, a social media post — is often described by several of these at once. Rather than attempting an exhaustive, mutually exclusive classification (which, in the current data landscape, is not realistically achievable), this reading follows a deliberately **simplified grouping**: most (not all) everyday data can be usefully thought of as either *structured* or *semi-/unstructured*.

#### Common Pitfall

The structured/unstructured split introduced below is a teaching simplification, not a strict, universally agreed-upon taxonomy. Some data (a time-stamped sensor log, a GPS trace) sits uncomfortably between the two categories, and reasonable practitioners sometimes classify the same dataset differently depending on how it will be used. Treat this grouping as a useful starting vocabulary, not as a rigid rule to apply mechanically.

### 3.2 Structured Data

**Structured data** is data that is organized according to a predefined, consistent format — typically a table with named columns, where every row shares the same fields. It is usually collected by filling out a form (either on paper or online), and it is typically stored in a database, a spreadsheet, or a plain-text tabular format such as `.csv` (comma-separated values).

| Everyday activity                          | The structured data it generates                                                                                     |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Opening a bank account                     | Name, date of birth, address, account type, initial deposit amount                                                   |
| Registering at a university                | Student ID, program, enrollment date, declared major                                                                 |
| Creating a Gmail, Amazon, or Azure account | Username, account creation date, subscription tier, linked payment method                                            |
| A visit to a doctor, or crossing a border  | Structured fields in a health record or immigration profile: vital signs, diagnosis codes, visa category, entry date |

A concrete, widely used example is the Kaggle [loan approval dataset](#), in which each row is one loan applicant, and each column is a fixed field: gender, marital status, number of dependents, education level, self-employment status, and applicant income, among others. Every applicant’s record has exactly the same columns, in exactly the same order — the defining feature of structured data.

### Key Idea

Structured data is easy to underestimate precisely because it is so familiar. Its real advantage is that a computer can immediately compute over it — sum a column, filter rows, join two tables — without first needing to figure out what the data even represents. As you will see in Sections 4–5, most of the effort in working with other data modalities is spent trying to reach this same, immediately computable, tabular form.

Not every structured (or structured-adjacent) dataset is a simple, standalone table, however. Some structured data is naturally a **sequence** or a **time series** — a record whose order matters and whose entries are related to their neighbors:

- **Standalone records:** a single genomic sample; a single day’s weather reading; a single map coordinate.
- **Sequences and series:** a stock’s daily closing price over a year; a credit card’s full transaction history; the sequence of GPS pings that reconstructs a delivery route; entries in a software system’s automatically generated log file.

This distinction matters because the two categories call for different analytical treatment: a standalone record can typically be analyzed on its own, while a sequence or series usually needs to be analyzed *in context* — an unusually large credit card charge is only suspicious relative to that same customer’s typical spending pattern, not in isolation.

### 3.3 Semi-/Unstructured Data

**Semi-/unstructured data** is data with no fixed, predefined tabular structure — it arrives as **free-form** content and is generally stored in a **data lake** or similar large-scale storage system designed to hold data before its structure has been fully determined. This is the modality of most content people would recognize as “media” rather than “records”:

- **Language (text):** a sequence of strings whose meaning (semantics) depends on word choice, order, and context — an email, a search query, a review.
- **Audio:** language combined with acoustic properties (pitch, tone, rhythm), covering both spoken language and music.
- **Image:** a visual representation of the world at a single instant.
- **Video:** a sequence of images (frames) over time, adding a temporal dimension to visual data.

As with structured data, some of this content is effectively standalone (a single photograph) while other content is inherently sequential (a video, a multi-turn conversation). **Social media data** is a particularly rich, hybrid example: a single post can combine free-form text, an attached image or video, and metadata capturing engagement (likes, shares) and even inferred emotional tone — illustrating just how thoroughly a single real-world artifact can blend modalities that a clean taxonomy would prefer to keep separate.

| Aspect                  | Structured data                                | Semi-/unstructured data                         |
|-------------------------|------------------------------------------------|-------------------------------------------------|
| Format                  | Fixed rows and columns                         | Free-form, variable format                      |
| Typical storage         | Database, spreadsheet, <code>.csv</code>       | Data lake / warehouse                           |
| Collected via           | Forms (manual or online)                       | Naturally produced media and communication      |
| Examples                | Loan applications, bank records, sensor tables | Photos, audio, video, free text, social posts   |
| Immediately computable? | Yes, largely                                   | No — typically needs encoding first (Section 4) |

### 3.4 Real-World Bridge: The Electronic Health Record

The following is an illustrative example, not drawn directly from the slides, chosen because it combines both categories in one familiar setting. A single hospital patient’s electronic health record (EHR) is rarely just one modality. It typically contains: *structured* data (lab values, vital signs, billing codes, medication dosages, all in fixed fields); *semi-structured* data (a nested list of prior diagnoses with dates, not a flat table but not free text either); and genuinely *unstructured* data (a physician’s free-text clinical notes describing the patient’s condition in ordinary language). A system like the Johns Hopkins TREWS early-warning tool [2], discussed in the companion KDD-process reading, depends primarily on the structured portion of this record (vital signs, lab results) precisely because structured data is the easiest modality to feed into a real-time monitoring algorithm — while the unstructured clinical notes, which often contain equally important information, require substantially more processing before they can be used the same way. This is a preview of a theme developed further in Section 4: not all modalities are equally ready for a data mining algorithm to consume, and a large part of real KDD work is closing that gap.

#### Practical Exercises

1. Classify each of the following as structured or semi-/unstructured, and justify any case where the classification feels uncomfortable or borderline: (a) a spreadsheet of monthly rent payments, (b) a customer service phone call recording, (c) a smartwatch’s minute-by-minute heart rate log, (d) a restaurant review posted online.
2. The reading distinguishes “standalone” records from “sequences/series.” For the credit card example, explain concretely why a single transaction, viewed alone, might look completely unremarkable, while the same transaction, viewed as part of a sequence, could indicate fraud.
3. Using the EHR example, describe one analytical question a hospital might want answered that requires *only* the structured portion of the record, and one question that would require the unstructured clinical notes as well.
4. Why do you think most large organizations store unstructured data in a “data lake” rather than forcing it into a traditional database table immediately upon arrival? Consider what would be lost, or how much extra work would be required, if every photo or audio file had to be converted into rows and columns before it could be stored at all.

## 4 Basic Data Types: Formalizing Data for Analysis

Section 3 grouped data by its everyday *modality* — what it looks like and how it is collected. This section introduces a different, complementary classification: the **basic data type** of an individual variable, which describes what kind of values it can take and what operations are meaningful on it. This distinction matters because a modality (say, “structured data”) can still contain columns of very different basic types, and choosing the wrong type for a given column is one of the most common early mistakes in a data mining project.

At the lowest level, most data falls into one of two basic types: **numerical** or **categorical**.

### 4.1 Numerical (Quantitative) Data

**Numerical data**, also known as **quantitative data**, represents quantities that can be added, averaged, and meaningfully compared using “greater than” or “less than.” It comes in two flavors, describing *where the number came from*:

- **Measurements**: a value obtained by measuring something on a continuous scale, such as a person’s height, weight, or blood pressure.
- **Counts**: a value obtained by counting discrete occurrences, such as the number of shares of stock a person owns, the number of teeth a dog has, or the number of pages in a favorite book.

Numerical data can also be classified along a second, closely related dimension — **discrete** versus **continuous**:

|                | Discrete                                                               | Continuous                                                                                                           |
|----------------|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Definition     | Values that can be counted and listed out explicitly (0, 1, 2, 3, ...) | Values that represent a measurement and cannot be fully listed, since any two values have another value between them |
| Typical origin | A count                                                                | A measurement                                                                                                        |
| Examples       | Number of stock shares, number of children, number of items in a cart  | Height, weight, IQ, blood pressure, temperature                                                                      |

#### Key Idea

Notice that “counts vs. measurements” and “discrete vs. continuous” describe almost the same underlying distinction from two angles: a count is almost always discrete, and a measurement is almost always continuous. It is useful to know both sets of terms, since you will encounter each in different textbooks and tools, but you do not need to treat them as two independent classification systems to memorize separately — in practice, knowing whether a number came from counting or measuring will usually tell you whether it is discrete or continuous as well.

### 4.2 Categorical (Qualitative) Data

**Categorical data** represents characteristics or qualities rather than quantities — a person’s gender, marital status, country of birth, or preferred movie genre. Arithmetic on categorical data (adding “married” and “single”) is meaningless, even when categories happen to be coded as numbers (a

point revisited in the pitfall box below). Categorical data splits into two further types, based on whether the categories have a meaningful order:

- **Ordinal data:** categories with a natural, meaningful order, such as a student’s letter grade (A, B, C, ...) or the days of the week. You can say one ordinal value is “higher” or “comes before” another, but not by how much in any numerically precise sense — the gap between an A and a B is not necessarily the same size as the gap between a B and a C.
- **Nominal data:** categories with no inherent order, such as a person’s gender, marital status, or country of birth. There is no meaningful sense in which one category is “more” or “less” than another.

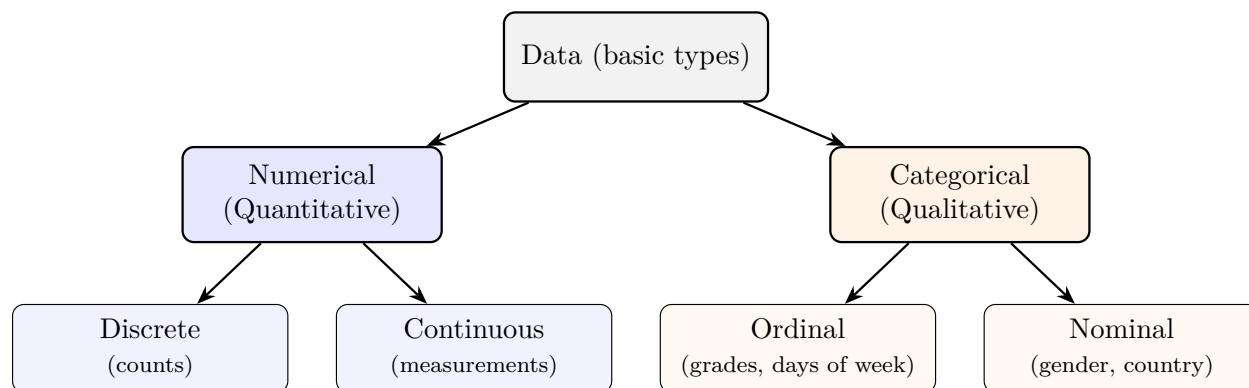


Figure 2: Basic data type taxonomy.

### Common Pitfall

**Numeric-looking data is not automatically numerical data.** *This nuance is added context, not drawn directly from the slides, but it is a common and consequential trap.* A U.S. zip code, a phone number, and a jersey number are all stored as digits, but none of them are numerical data in the sense defined above: averaging two zip codes produces a meaningless result, and “zip code 90210” is not twice “zip code 45120.” These are **nominal categorical** variables that merely happen to use numerals as their category labels. Before running any statistic on a column, ask what the numbers actually *represent* — not just what data type the computer thinks they are.

*Illustrative example (not from the slides):* a streaming platform’s customer-satisfaction survey asks users to rate their experience from 1 (very dissatisfied) to 5 (very satisfied). This is **ordinal**, not continuous numerical, data: a rating of 4 is better than a rating of 2, but it is not meaningful to claim it represents “twice as much” satisfaction, and the gap between a 1 and a 2 is not guaranteed to feel the same to respondents as the gap between a 4 and a 5. Treating Likert-scale survey data as if it were ordinary continuous numerical data (for example, by casually averaging it and interpreting the average the way you would average height) is a common source of subtly overstated conclusions in customer analytics.

## Practical Exercises

1. Classify each of the following as numerical (discrete or continuous) or categorical (ordinal or nominal): (a) a book's page count, (b) a hotel's star rating (1–5 stars), (c) a customer's preferred payment method, (d) a runner's marathon finishing time, (e) a product's size label (Small/Medium/Large).
2. Explain, in your own words, why the “discrete vs. continuous” distinction and the “counts vs. measurements” distinction are so closely related. Can you construct a plausible (even if unusual) example of a count that behaves more like continuous data, or a measurement that behaves more like discrete data?
3. A dataset stores a customer's loyalty tier as the numbers 1, 2, and 3 (instead of “Bronze,” “Silver,” “Gold”). Is this column numerical or categorical? Does your answer change depending on whether the tiers are ordered by spending threshold?
4. Revisit the loan approval dataset from Section 3.2. For each of the following columns, state its basic data type: `ApplicantIncome`, `Education` (Graduate/Not Graduate), `Dependents` (0, 1, 2, 3+), `Self_Employed` (Yes/No).

## 5 From Modality to Machine-Readable Data: Encoding Images as Numbers

Section 3 introduced image data as one modality among several unstructured types. This section asks a more concrete question: *if a data mining algorithm ultimately only understands numbers, how does a picture — which has no obvious rows and columns — ever become something an algorithm can use?* The answer is **encoding**: a systematic, reversible way of representing a modality as numeric data.

### 5.1 Why Encoding Matters

Every unstructured modality discussed in Section 3.3 needs its own encoding scheme: text is typically encoded as sequences of numeric token identifiers, audio as sequences of amplitude samples, and images as grids of numeric pixel values. This reading works through the image case in detail, both because it is the most visually intuitive and because it illustrates a pattern (raw modality → numeric grid → flattened vector) that recurs, with variations, across every other modality you will encounter later in this course.

### 5.2 Binary, Grayscale, and Color Images

An image, at the level a computer actually stores it, is a grid of **pixels**, and each pixel is represented by one or more numbers. The number of values needed per pixel, and the range those values can take, depends on the type of image:

| Image type         | What each pixel stores                            | Value range          | Channels per pixel |
|--------------------|---------------------------------------------------|----------------------|--------------------|
| Binary             | Black or white only                               | {0, 1}               | 1                  |
| Grayscale<br>(B/W) | Shade of gray (intensity)                         | [0, 255]             | 1                  |
| Color (RGB)        | Intensity of red, green, and blue light, combined | [0, 255] per channel | 3                  |

A small  $3 \times 3$  patch of each image type makes the idea concrete:

*Binary patch*

$$\begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{pmatrix}$$

*Grayscale patch*

$$\begin{pmatrix} 129 & 115 & 121 \\ 115 & 109 & 120 \\ 136 & 162 & 173 \end{pmatrix}$$

*Color patch (one pixel)*

$$R = 165, G = 187, B = 209$$

A binary image needs only a single bit per pixel (on or off), which is why it is the simplest and smallest representation — but also the least expressive, since it cannot represent shading or color at all. A grayscale image needs one number per pixel, typically ranging from 0 (black) to 255 (white), giving 256 distinguishable shades of gray. A color image needs *three* numbers per pixel — one each for red, green, and blue intensity — which combine to reproduce the full range of colors a screen can display.

### Key Idea

The value range  $[0, 255]$  is not arbitrary: it is exactly the range representable by a single **byte** (8 bits), since  $2^8 = 256$  distinct values. This is why image intensities are so often described in this specific range — it reflects a natural, efficient match to how computers store data at the hardware level, not merely a historical convention.

## 5.3 The Vector-Size Formula

Once an image is represented as a grid of numbers, it can be **flattened** into a single, long list of numbers — a **feature vector** — which is the format most data mining and machine learning algorithms expect as input. The size of that vector follows directly from the image's dimensions and number of channels:

$$\text{vector length} = H \times W \times C$$

where  $H$  is the image height in pixels,  $W$  is the image width in pixels, and  $C$  is the number of channels per pixel (1 for binary or grayscale images, 3 for color/RGB images).

### Tip

**Worked example.** What is the vector size of a  $40 \times 50$  RGB color image? Here  $H = 40$ ,  $W = 50$ , and  $C = 3$  (red, green, blue), so:

$$\text{vector length} = 40 \times 50 \times 3 = 6,000$$

The same image stored as grayscale instead of color would need only  $40 \times 50 \times 1 = 2,000$  values — a factor of three smaller, entirely because grayscale needs only one channel per pixel instead of three.

### Common Pitfall

This formula is also a preview of a challenge you will meet repeatedly in this course: image, audio, and video data produce *very* long feature vectors even for modestly sized inputs. A single, unremarkable  $200 \times 200$  color photograph — small by modern camera standards — already flattens into a vector of  $200 \times 200 \times 3 = 120,000$  numbers. Managing this scale (dimensionality) efficiently is a recurring concern in data mining that begins with the simple arithmetic in this section.

*Illustrative extension (not from the slides):* in medical imaging, a diagnostic system for detecting pneumonia from a chest X-ray typically encodes each grayscale X-ray image as exactly this kind of flattened numeric vector before any classification algorithm ever sees it. The algorithm has no innate notion of “lungs” or “shadows”; it only ever operates on the numeric intensity values that Section 5.2 describes, which is precisely why the encoding step — turning a modality (an image) into numbers a machine can use — is not a minor technical detail but a prerequisite for the entire analysis.

### Practical Exercises

1. Compute the flattened vector size for: (a) a  $28 \times 28$  binary image, (b) a  $28 \times 28$  grayscale image, (c) a  $28 \times 28$  RGB color image. What single factor accounts for all of the differences between your three answers?
2. A grayscale image and a binary image of the same resolution both use 1 channel per pixel, yet a binary image is described in this section as needing far less storage per pixel than a grayscale image. Reconcile this: what is different between the two, if not the channel count used in the vector-length formula?
3. Explain, in your own words, why an algorithm trained on flattened image vectors has “no innate notion of lungs or shadows.” What does this suggest about why simply having enough numbers is not the same as having usable information?
4. A video is a sequence of image frames (Section 3.3). Using the vector-length formula from this section as a starting point, propose an expression for the total amount of numeric data needed to represent one second of  $200 \times 200$  RGB color video recorded at 30 frames per second.

## 6 Connecting the Concepts: From Raw Data to Knowledge Discovery

Sections 2 through 5 can feel like three separate topics — a growth chart, a taxonomy of everyday data, and an arithmetic formula for image vectors — but they describe one continuous story about how the data mining process introduced in the companion KDD reading actually gets started.

- **Growth (Section 2)** explains *why* this all matters at the scale it does: there is simply too much data, arriving too quickly, for manual review to keep up, which is what makes an automated, structured process like KDD necessary in the first place.
- **Modalities (Section 3)** describe *what* that data actually looks like when it arrives — and crucially, most of it does *not* arrive already in the clean, tabular form a mining algorithm needs.

- **Basic types and encoding** (Sections 4–5) describe *how* raw data, whatever its modality, gets formalized into a representation — numerical, categorical, or a flattened numeric vector — that the **Transformation** stage of the KDD process (see the companion reading) can actually operate on.

### Key Idea

Put in the vocabulary of the KDD process from the companion reading: recognizing a variable’s basic data type, and knowing how to encode an unstructured modality as numbers, are exactly the kinds of decisions made during **Preprocessing** and **Transformation**. Get them wrong — treat a zip code as numerical, or feed an algorithm raw image files instead of a numeric vector — and every later stage inherits the mistake.

*Capstone illustrative example (not from the slides).* Consider a streaming video service, whose data mining team wants to improve content recommendations. A single subscriber generates all of the concepts in this reading at once: **structured** signup data (name, billing tier, join date — Section 3.2); a **sequence** of viewing events, timestamped and ordered (Section 3.2’s “sequences and series” distinction); **unstructured** text reviews and unstructured audio in the content itself (Section 3.3); **categorical** genre labels that are nominal (comedy, drama, documentary) alongside a **numerical** view count and an **ordinal** star rating (Section 4); and thumbnail **images** for every title in the catalog, each of which must be encoded into a numeric vector (Section 5) before any recommendation algorithm can use visual similarity as a signal at all. No single stage of this reading, on its own, is enough to build a recommendation system — but together, they describe exactly the raw material a data mining pipeline has to work with before mining, in the narrower sense, can even begin. This mirrors, at a much larger scale, the same structure as the bookstore order-log capstone in the companion KDD reading: a mix of numeric, categorical, and (there, absent) unstructured fields, all needing to pass through Selection, Preprocessing, and Transformation before any pattern can be mined from them.

## 7 Common Misconceptions and Important Nuances

### Common Pitfall

**“Structured vs. unstructured” is a convenience, not a law of nature.** As Section 3.1 states explicitly, some data (time-stamped logs, GPS traces) does not sit cleanly in either category. Do not lose points on an exam, or make a bad modeling decision in practice, by forcing an awkward real dataset into whichever category is more convenient rather than describing it accurately.

### Common Pitfall

**A “sequence” is not automatically an unstructured modality.** Stock prices and credit card histories (Section 3.2) are structured *and* sequential at the same time. Sequence/stand-alone and structured/unstructured are two independent distinctions, not two names for the same thing.

### Common Pitfall

**Discrete numerical data is still numerical, not categorical, even though it can be listed out.** A count of children (0, 1, 2, 3, ...) is discrete, but it remains meaningfully ordered and arithmetically combinable (you can compute an average number of children per household). Do not confuse “discrete” (a property of numerical data) with “categorical” (a fundamentally different basic type).

### Common Pitfall

**Ordinal data supports ordering comparisons but not, in general, arithmetic ones.** You can correctly say a letter grade of A “beats” a B, or that Friday “comes after” Wednesday. You cannot safely assume the size of the gap between adjacent ordinal categories is constant — treating ordinal survey ratings as if they were evenly spaced continuous numbers (Section 4.2) is a common source of overconfident conclusions in customer analytics.

## 8 Key Takeaways for CIS 635

1. **Data volume has grown exponentially, and recently.** The shift from analog to digital defaults, combined with social media, IoT sensors, and generative AI, produced a data landscape that manual, ad hoc analysis cannot keep up with — which is precisely why a structured discipline like data science, and a structured process like KDD, are necessary.
2. **Structured vs. semi-/unstructured is a useful simplification, not a rigid taxonomy.** Most everyday data can be usefully grouped this way, but the grouping is explicitly incomplete, and some data resists clean classification.
3. **Sequence/standalone is a separate distinction from structured/unstructured.** A stock’s price history is structured and sequential; a single photograph is unstructured and standalone — all four combinations exist.
4. **Every variable has a basic data type — numerical (discrete or continuous) or categorical (ordinal or nominal) — and getting it wrong has real consequences.** Numeric-looking codes (zip codes, jersey numbers) are frequently categorical in disguise; ordinal ratings are frequently mistaken for evenly spaced continuous numbers.
5. **Unstructured modalities must be encoded as numbers before any algorithm can use them.** For images, this means representing each pixel with 1 channel (binary/grayscale) or 3 channels (RGB color), then flattening the result into a feature vector of length  $H \times W \times C$ .
6. **These three ideas — growth, modalities, and types/encoding — describe the raw material the KDD process (from the companion reading) is built to handle.** Recognizing a variable’s type and knowing how to encode a modality are concrete instances of the Preprocessing and Transformation stages you have already studied.

## 9 Discussion Questions

1. Section 2 argues that data science “industrialized,” rather than invented, centuries-old statistical ideas. Do you find this framing convincing? What, if anything, is genuinely new about

modern data science beyond scale and computing power?

2. The reading repeatedly emphasizes that its classification schemes (structured/unstructured, discrete/continuous, ordinal/nominal) are useful simplifications with known exceptions. Why do you think a course would teach an admittedly imperfect taxonomy at all, rather than waiting until a fully precise one can be given?
3. Compare the electronic health record example (Section 3.4) and the streaming-service capstone (Section 6). Both combine multiple modalities and data types in one real-world entity. What practical challenges do you think a data mining team would face that are common to both scenarios, and what challenges would be specific to just one of them?
4. The vector-length formula in Section 5.3 shows that image data becomes very large, very quickly, as resolution increases. Speculate about one strategy a real system might use to keep this manageable, and explain, at a conceptual level, what information that strategy might sacrifice in exchange for a smaller vector.
5. This reading’s companion KDD-process reading warned that widely repeated data mining anecdotes (such as the “beer and diapers” story) are not always accurate. Are the data-growth figures in Section 2 the kind of claim that deserves the same scrutiny, or are they on more solid footing? What would you want to know about how Statista produced its forecast before citing it yourself?

## References

## References

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